FOMC Event Driven Trading Technical Writeup (Draft)

Diego Alvarez

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1 Introduction

This write-up develops an event-driven trading model for Treasury futures centered around FOMC announcement days. The framework is primarily motivated by the work of Peng and Pan in Pre-FOMC Announcement Drift,[1] which examines the behavior of Treasury yields leading up to FOMC meetings. While their research focuses on identifying a risk premium embedded in pre-FOMC interest rate movements, the same intuition can be extended to the Treasury futures market. This study replicates those ideas within the futures space and further develops them into a straightforward backtesting framework that can later be expanded and refined.

This paper is organized as follows. It begins with a discussion of the data, followed by research and validation of the findings presented by Peng and Pan, along with an analysis of the underlying baseline behavior. The paper then proceeds to the construction of several backtesting frameworks and concludes with an evaluation of the resulting performance and findings.

2 Data

The data consists of Treasury futures that are collected via Bloomberg. The following is a table of futures used with their respective start and end dates. These are roll-adjusted futures that are rolled by Bloomberg relative to their delivery date and on a ratio basis. Although Treasury futures should be rolled via their basis points on an open interest crossover, this methodology is standard within the GFUT method and will suffice.

Table 1: Treasury Futures Data

Security	Description	Start Date	End Date
FV1 Comdty	5y Treasury Note	2004-05-24	2024-09-20
TU1 Comdty	2y Treasury Note	2004-05-24	2024-09-20
TY1 Comdty	10y Treasury Note	2004-05-24	2024-09-20
US1 Comdty	20y Treasury Bond	2004-05-24	2024-09-20
UXY1 Comdty	10y Treasury Note	2016-01-13	2024-09-20
WN1 Comdty	30y Treasury Bond	2010-01-13	2024-09-20

Generic Treasury yield data was obtained from the St. Louis Federal Reserve Bank via Federal Reserve Economic Data (FRED), focusing on maturities greater than one year. Since the primary focus of this project is return behavior around FOMC announcement days, additional data on the Federal Funds Target Rate (FDTR) release dates was sourced from Bloomberg LP.

NLP-based sentiment data is taken from Bloomberg Economics sentiment indicators, some of which are derived using natural language processing techniques. These measures are computed internally by

1. Qing Peng and Jun Pan, "The Pre-FOMC Drift and the Secular Decline in Long-Term Interest Rates," Available at SSRN: https://ssrn.com/abstract=4764451 or http://dx.doi.org/10.2139/ssrn.4764451, March 2024,

Bloomberg and accessed through the Bloomberg Terminal.

Security	Description
BENLPFPH	Bloomberg Economics Federal Reserve Sentiment Hawk Probability Hike Next Meeting
BENLPFED	Bloomberg Economics Federal Reserve Sentiment Natural Language Processing
APUSISGF	Bloomberg Economics Decomposition of US Ten year Inflation Swap - Risk Sentiment
APUSSPGF	Bloomberg Economics Decomposition of US S&P 500 - Risk Sentiment News
APUSTYGF	Bloomberg Economics Decomposition of US 10 Year Bond Yield - Risk Sentiment News

3 Background

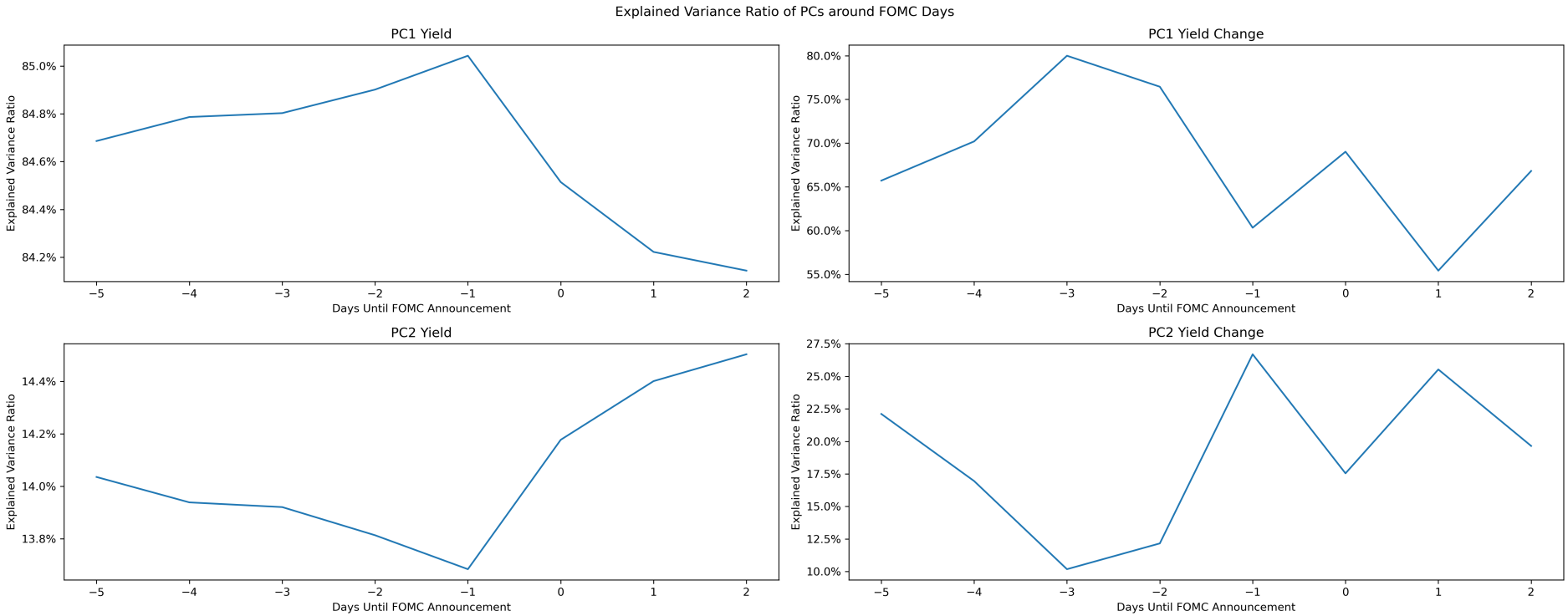
The analysis focuses on changes in Treasury yields around FOMC announcements, with particular attention to their structure through a principal component analysis (PCA) framework. The objective is to examine whether there is a dominant secular driver of rate movements in the vicinity of FOMC events, which would be reflected in an elevated proportion of explained variance captured by the first principal component.

A secondary consideration is that Treasury futures are being traded in a pervasive manner; therefore, establishing the presence of a strong common factor is important for motivating and supporting the construction of the subsequent backtesting framework.

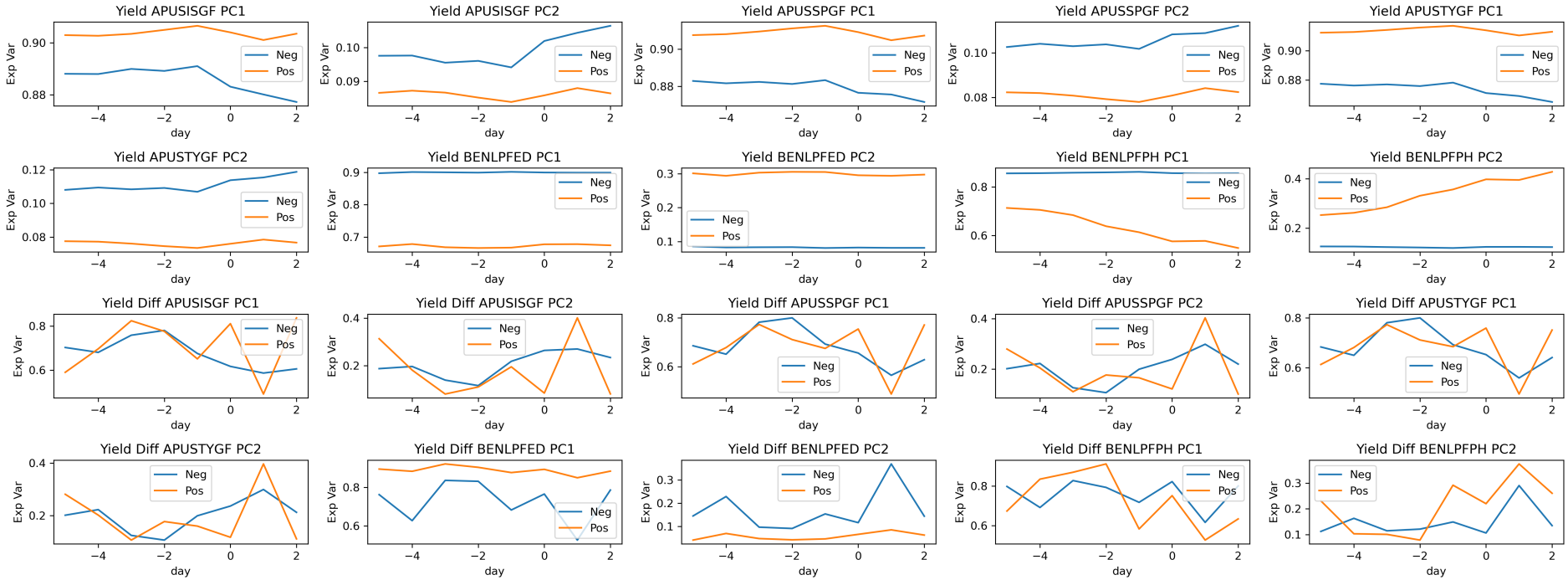
The following page presents the explained variance of the first principal component for Treasury yields and their daily changes. A noticeable increase in explained variance is observed leading up to FOMC announcements, followed by a decline after the announcement date. To demonstrate that the explained variance of the principal components around FOMC dates is elevated relative to normal periods, a PCA is performed on Treasury yield changes and compared against the full-sample explained variance. The results in the table below show that the explained variance around FOMC events is meaningfully higher than that of the overall sample.

	PCA Period	Spef. Day		Full Period	
		Raw Yield	Yield Changes	Raw Yield	Yield Changes
Days	PC				
-5	PC1	0.847	0.657	0.858	0.679
	PC2	0.140	0.221	0.129	0.178
-4	PC1	0.848	0.702	0.858	0.679
	PC2	0.139	0.169	0.129	0.178
-3	PC1	0.848	0.800	0.858	0.679
	PC2	0.139	0.102	0.129	0.178
-2	PC1	0.849	0.765	0.858	0.679
	PC2	0.138	0.122	0.129	0.178
-1	PC1	0.850	0.603	0.858	0.679
	PC2	0.137	0.267	0.129	0.178
0	PC1	0.845	0.690	0.858	0.679
	PC2	0.142	0.175	0.129	0.178
1	PC1	0.842	0.554	0.858	0.679
	PC2	0.144	0.255	0.129	0.178
2	PC1	0.841	0.668	0.858	0.679
	PC2	0.145	0.196	0.129	0.178

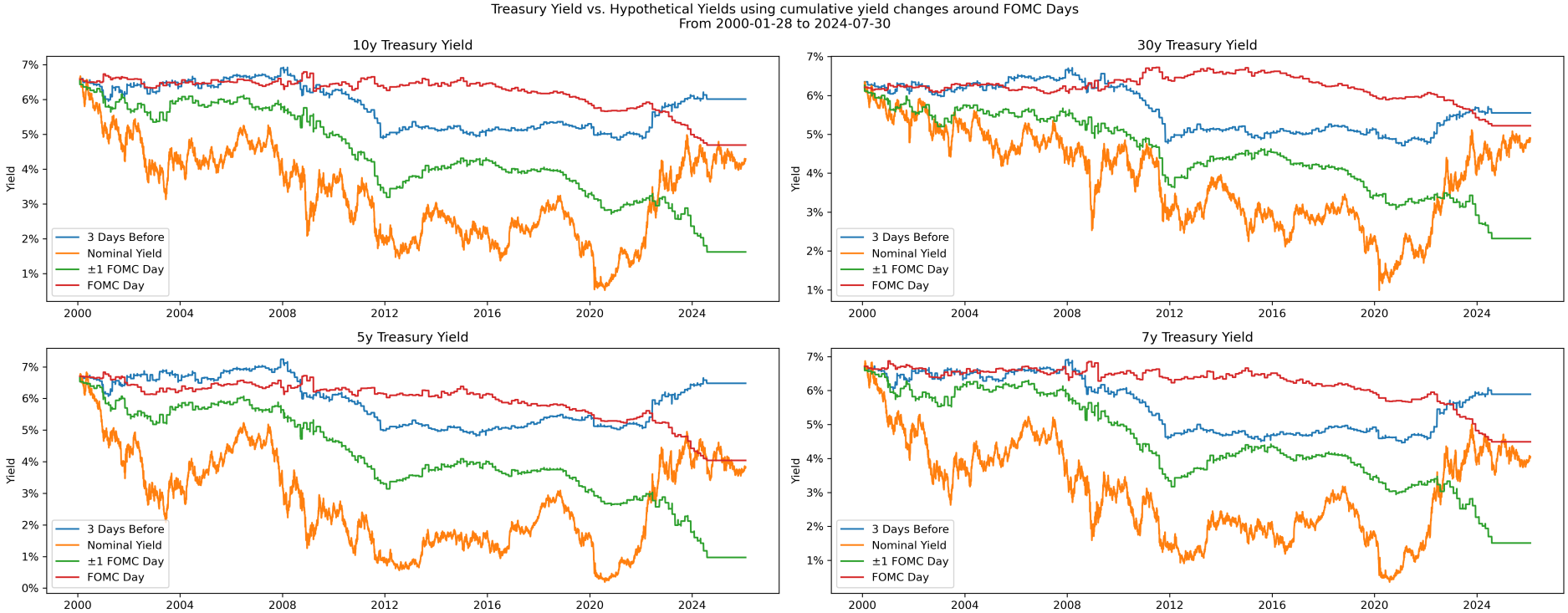
A 10-day exponentially weighted z-score is then calculated for the NLP sentiment measures. The sample is subsequently conditioned on whether the z-score is positive or negative, and the explained variance from the PCA is recalculated under each regime. The following page presents the conditioned explained variance results.


Explained Variance Ratio of PCs around FOMC Days
PC1 Yield
PC1 Yield Change
PC2 Yield
PC2 Yield Change
Explained Variance Ratio
Days Until FOMC Announcement
85.0%
84.8%
84.6%
84.4%
84.2%
80.0%
75.0%
70.0%
65.0%
60.0%
55.0%
14.4%
14.2%
14.0%
13.8%
27.5%
25.0%
22.5%
20.0%
17.5%
15.0%
12.5%
10.0%
−5
−4
−3
−2
−1
0
1
2

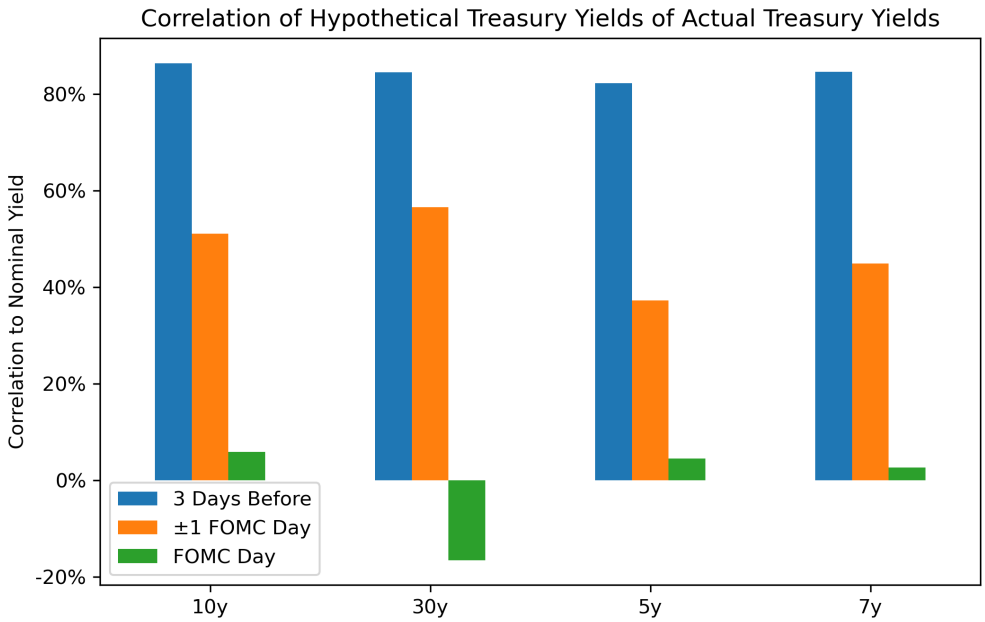
Explained Variance Ratios of First 2 Principal Components conditioned on sentiment factors based on positive or negative 10d Z-Score
From 1962-01-03 to 2026-02-02



The PCA analysis provides considerable evidence that Treasury yields are predominantly driven by FOMC-related activity during announcement periods. Using this framework, hypothetical yield paths can be constructed from yield changes around FOMC dates. The comparison of these hypothetical returns is presented below.



The correlation of the hypothetical yield to the real yields is quite large.



4 Naively Backtest

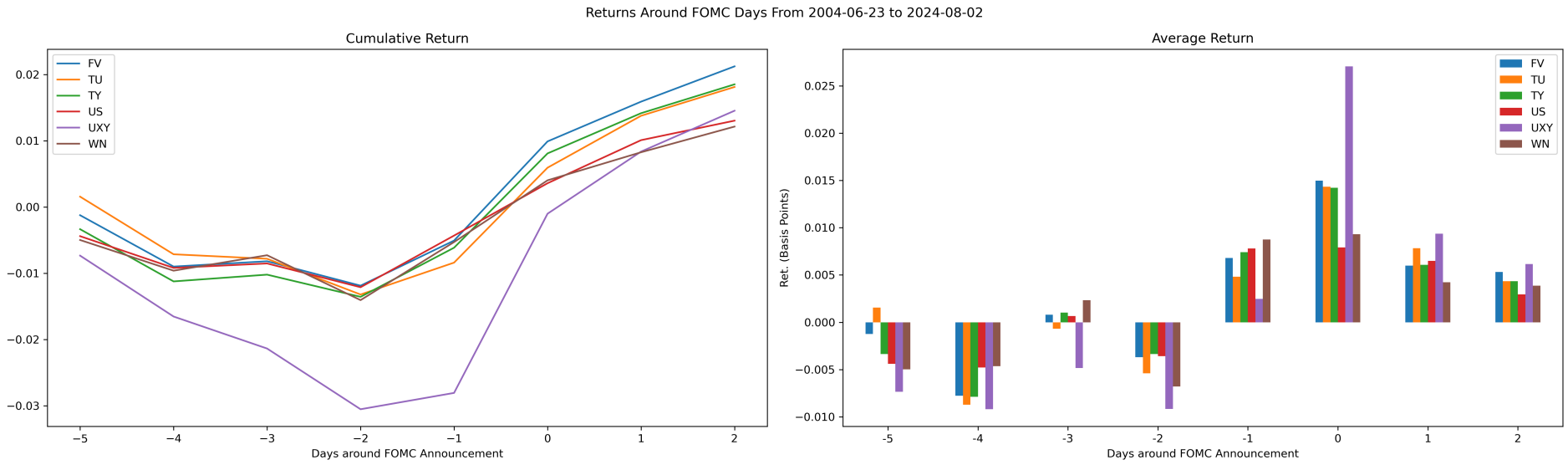
This section examines a set of backtests conditioned on sentiment signals. The analysis uses the most recent NLP-derived sentiment observation prior to each event to define the conditioning regime. Returns are then evaluated across these sentiment states to assess how performance varies under different sentiment conditions. All futures returns are expressed in basis points using standard duration-based scaling.

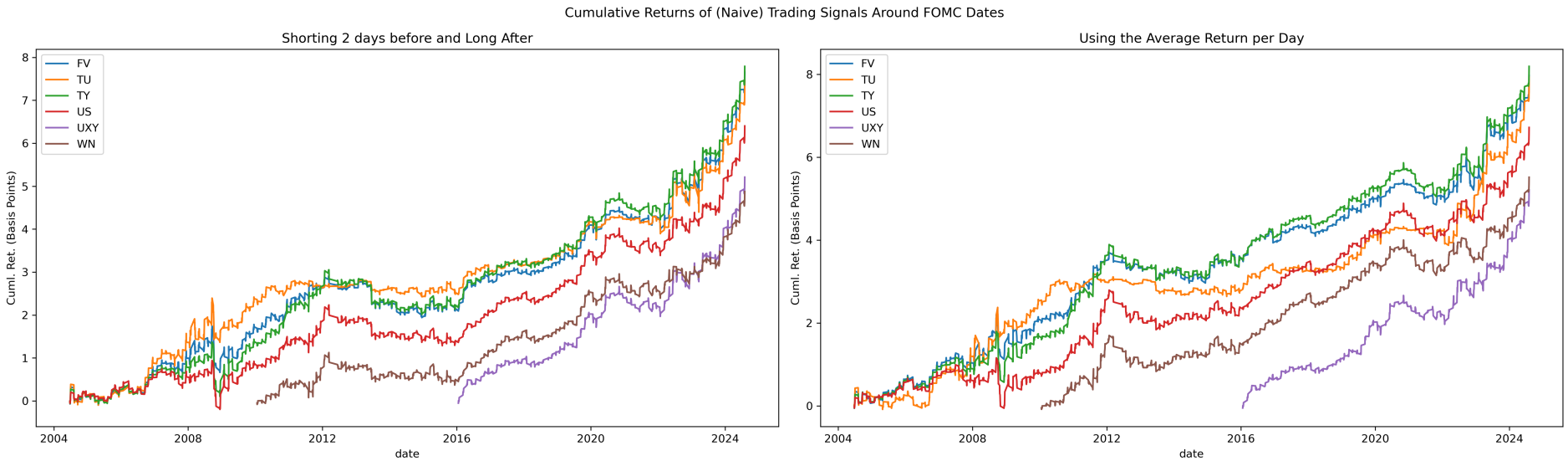
A simple approach is first implemented by examining Treasury returns around FOMC announcement days and using the sign of the announcement-day return as a trading signal for Treasury futures. The framework is initially evaluated in-sample and subsequently extended to assess out-of-sample performance.

The second page presents a playback of futures returns around FOMC announcements. The analysis considers both cumulative returns and daily returns. Two backtests are constructed: the first takes a long position starting two days before the meeting, motivated by the cumulative returns pattern; the second trades based on the average return profile, informed by the bar chart on the right-hand side.

The following page presents the backtest results for the two trading strategies. Both strategies generate reasonable performance. A table summarizing their Sharpe ratios is shown below, with all Sharpe ratios reported on an annualized basis. For comparison, both the full-sample Treasury Sharpe and the Sharpe measured around FOMC periods are included. It is evident that both the average return strategy and the naive strategy outperform the benchmarks.

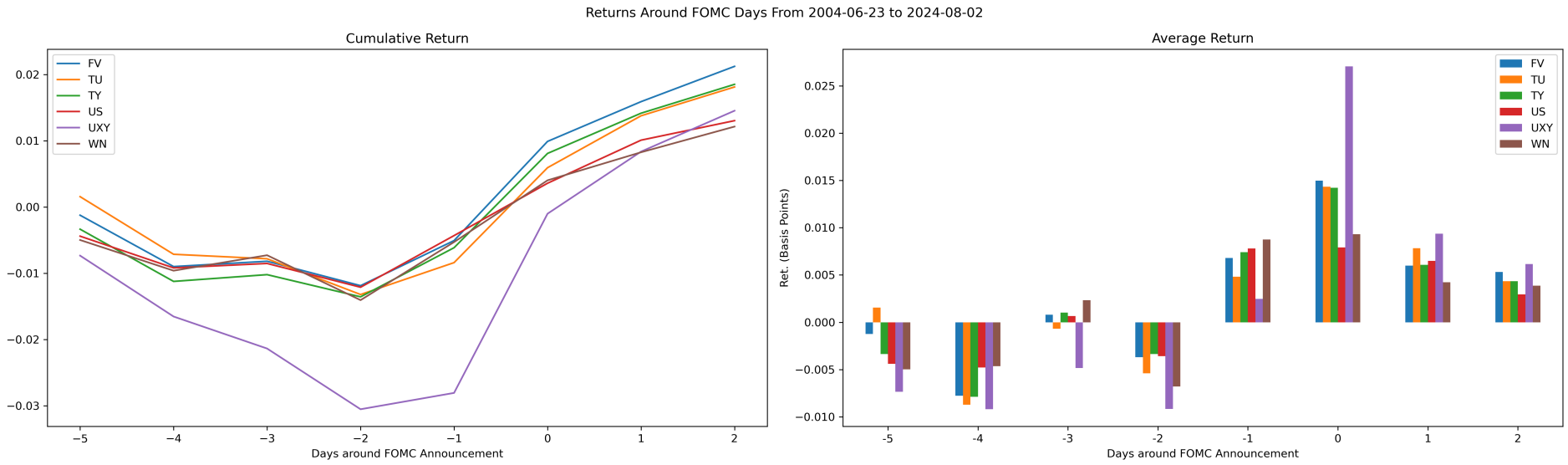
security	Avg. Ret. Strat	Naive Strat	Sharpe around FOMC	Full Hist Sharpe
FV	1.623	1.557	0.775	0.304
TU	1.606	1.512	0.645	0.282
TY	1.708	1.623	0.701	0.279
US	1.548	1.474	0.574	0.166
UXY	2.750	2.750	0.519	-0.215
WN	1.799	1.587	0.540	0.045
Avg	1.839	1.751	0.626	0.144


Returns Around FOMC Days From 2004-06-23 to 2024-08-02
Cumulative Return
FV
TU
TY
US
UXY
WN
0.02
0.01
0.00
−0.01
−0.02
−0.03
−5
−4
−3
−2
−1
0
1
2
Days around FOMC Announcement
Average Return
Ret. (Basis Points)
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
-5
-4
-3
-2
-1
0
1
2
Days around FOMC Announcement


Cumulative Returns of (Naive) Trading Signals Around FOMC Dates
Shorting 2 days before and Long After
Using the Average Return per Day
FV
TU
TY
US
UXY
WN
Cuml. Ret. (Basis Points)
date
0
1
2
3
4
5
6
7
8
2004
2008
2012
2016
2020
2024

The results change when moving to an out-of-sample setting. The first strategy is difficult to replicate out-of-sample, as it relies on continuously observing returns and manually determining the trading direction. In contrast, the second strategy can be implemented more systematically by computing an expanding average for each day. The table below reports Sharpe ratios for both in-sample and out-of-sample performance. In this case, nearly a full point of Sharpe is lost, effectively eliminating most of the strategy's risk-adjusted return. The following page presents a comparison of in-sample and out-of-sample returns.

security	In-Sample	Out-of-Sample	Sharpe Lost
FV	1.623	0.808	0.815
TU	1.606	0.491	1.115
TY	1.708	0.691	1.017
US	1.548	0.361	1.186
UXY	2.750	2.040	0.710
WN	1.799	0.559	1.240


Returns Around FOMC Days From 2004-06-23 to 2024-08-02
Cumulative Return
FV
TU
TY
US
UXY
WN
0.02
0.01
0.00
−0.01
−0.02
−0.03
−5
−4
−3
−2
−1
0
1
2
Days around FOMC Announcement
Average Return
Ret. (Basis Points)
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
−0.010
-5
-4
-3
-2
-1
0
1
2
Days around FOMC Announcement

References

Peng, Qing, and Jun Pan. "The Pre-FOMC Drift and the Secular Decline in Long-Term Interest Rates." Available at SSRN: https://ssrn.com/abstract=4764451 or http://dx.doi.org/10.2139/ssrn.4764451, March 2024.